MODULE 1 · LECTURE 19

Introduction to Artificial Intelligence: Definition, Background and Applications

What artificial intelligence is and how the field developed, its main branches, intelligent agents, how machines learn from data and how their results are judged, and where AI is being applied in agriculture — with its limits and responsibilities.

Learning objectives

After this lecture you should be able to:

1. Define artificial intelligence and compare the four approaches to defining it.
2. Outline the history of AI from the 1950s to today.
3. Describe the main branches of AI and the relationship between AI, machine learning and deep learning.
4. Describe an intelligent agent and specify it with PEAS.
5. Explain the types of machine learning and the supervised-learning workflow, and work through a k-nearest-neighbour example.
6. Evaluate a classifier with accuracy, precision, recall and F1 score.
7. Survey applications of AI in agriculture and discuss its limitations and responsible use.

Scope: Lectures 1–18 covered computers, data, the Internet, models and decision support; Lecture 17 introduced expert systems, an early form of AI. This lecture opens the AI part of the course. The Turing test and control strategies follow in Lecture 20, search in Lectures 21–22, and AI applications in Lectures 23–28.

1. What is artificial intelligence?

Artificial intelligence (AI) is the field that builds machines able to perform tasks that normally require human intelligence — perceiving, learning, reasoning, planning, communicating and acting. The OECD defines an **AI system** as "a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments", adding that different AI systems vary in their levels of autonomy and adaptiveness after deployment (Grobelnik et al., 2024).

Four approaches to defining AI

	Like humans	Rationally (doing the right thing)
Thinking	**Thinking humanly** — model how people think (cognitive science)	**Thinking rationally** — reason correctly by the laws of logic
Acting	**Acting humanly** — behave so that people cannot tell it from a human (the Turing test, Lecture 20)	**Acting rationally** — act to achieve the best expected outcome (the rational-agent approach)

Most modern AI follows the **rational-agent** approach: build systems that do the right thing given what they perceive and know (Russell & Norvig, 2020). All AI in use today is **narrow AI**, built for specific tasks such as recognising leaf diseases or answering scheme queries; **general AI**, matching humans across all tasks, does not exist.

2. Background: a brief history

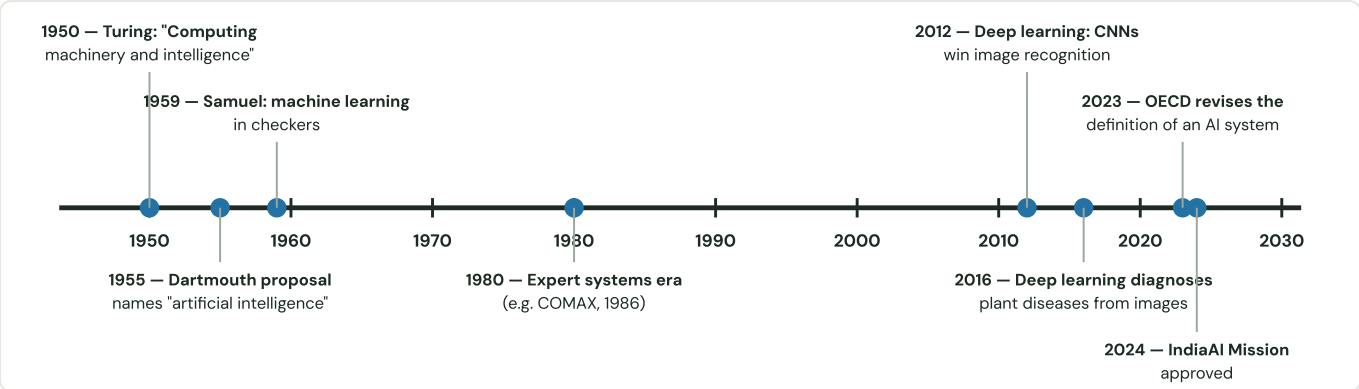


Figure 1. Milestones in the history of AI mentioned in this lecture. Original diagram.

- **1950** — Alan Turing asked whether machines can think and proposed an "imitation game" to test it (Turing, 1950; Lecture 20).
- **1955–56** — McCarthy, Minsky, Rochester and Shannon proposed a two-month, ten-man study of artificial intelligence at Dartmouth College for the summer of 1956, "on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it" (McCarthy et al., 1955). The field takes its name from this proposal.
- **1959** — Arthur Samuel's checkers program improved its play from experience — an early demonstration of machine learning (Samuel, 1959).
- **1960s–1980s** — symbolic AI: search, logic and knowledge-based expert systems such as COMAX for cotton (Lecture 17). Periods of over-optimism were followed by cuts in funding, often called "AI winters".
- **1990s–2000s** — statistical machine learning from growing data sets.
- **2012 onwards** — deep learning: a large convolutional neural network classified 1.2 million ImageNet images into 1,000 classes far better than earlier methods (Krizhevsky et al., 2017). Deep learning lets models learn layers of features directly from raw data (LeCun et al., 2015).
- **2020s** — generative AI and large language models; governments define and regulate AI systems (OECD definition revised in 2023) and invest in national AI capacity (IndiaAI Mission, 2024).

3. Branches of AI

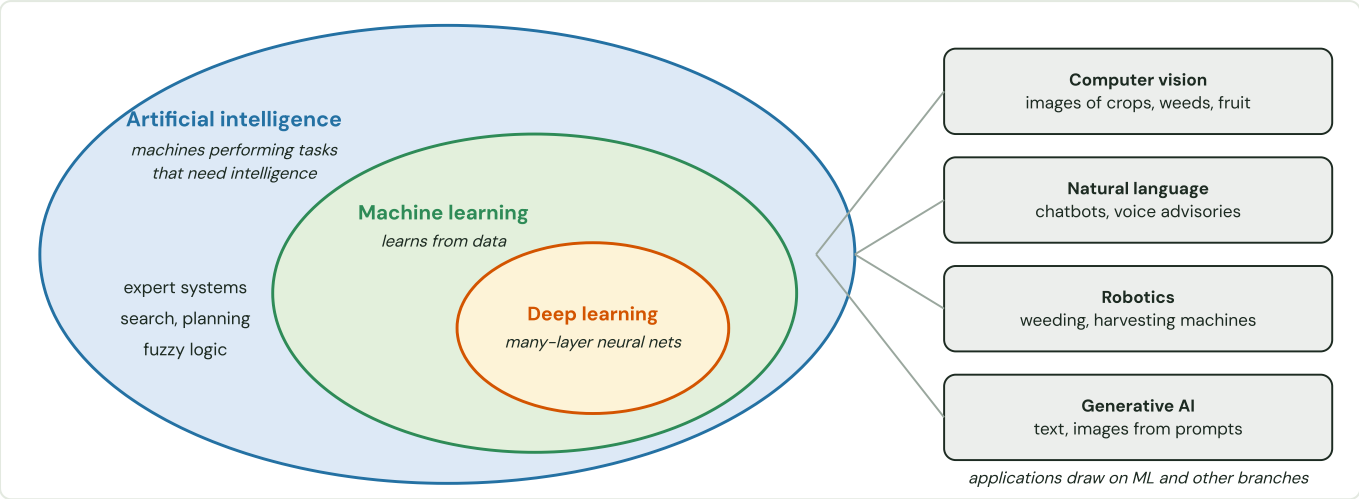


Figure 2. Machine learning is a part of AI, and deep learning a part of machine learning; applications draw on several branches. Original diagram.

Branch	What it does	Agricultural example
Machine learning	Learns patterns from data to predict or classify	Yield prediction from weather, soil and satellite data
Deep learning	Neural networks with many layers learn features from raw data	Leaf-disease recognition from photos
Computer vision	Interprets images and video	Weed detection for targeted spraying (Lectures 25, 27)
Natural language processing and speech	Understands and produces human language	Multilingual farmer chatbots and voice advisories
Knowledge-based systems	Reason with rules from experts	Expert systems for crop management (Lecture 17)
Search and planning	Find sequences of actions to reach goals	Path planning for field robots (Lectures 21–22)
Robotics	Sense and act in the physical world	Autonomous weeding and harvesting (Lecture 27)
Fuzzy logic	Reasons with degrees of truth	Controllers for irrigation and greenhouses

4. Intelligent agents

An **agent** is anything that perceives its environment through **sensors** and acts on it through **actuators**; a **rational agent** chooses the action expected to give the best performance (Russell & Norvig, 2020). An agent's task is specified by **PEAS** — **P**erformance measure, **E**nvironment, **A**ctuators and **S**ensors.

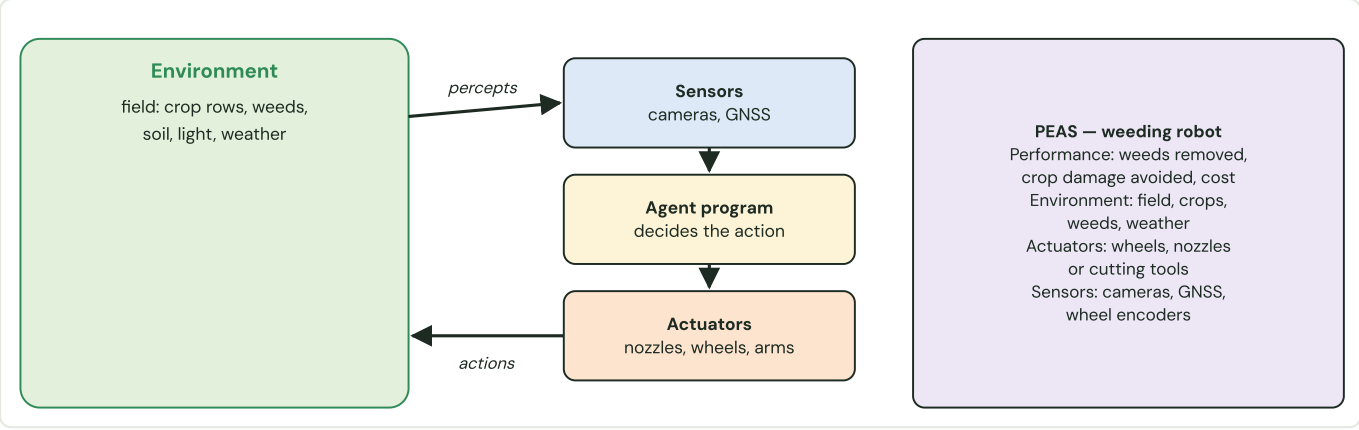


Figure 3. An agent interacting with its environment, with the PEAS description of a weeding robot. Original diagram.

5. How machines learn

Machine learning (ML) is the part of AI in which a system improves at a task by learning from data or experience instead of being programmed with every rule — the idea Samuel (1959) demonstrated with checkers.

Type	Learns from	Output	Agricultural example
Supervised learning	Examples with the correct answers (labels)	Classification (classes) or regression (numbers)	Healthy/diseased leaf; yield in t/ha
Unsupervised learning	Data without labels	Groups (clusters) or patterns	Grouping fields into management zones
Reinforcement learning	Rewards and penalties from acting	A policy — what to do in each situation	Learning irrigation or greenhouse-control decisions

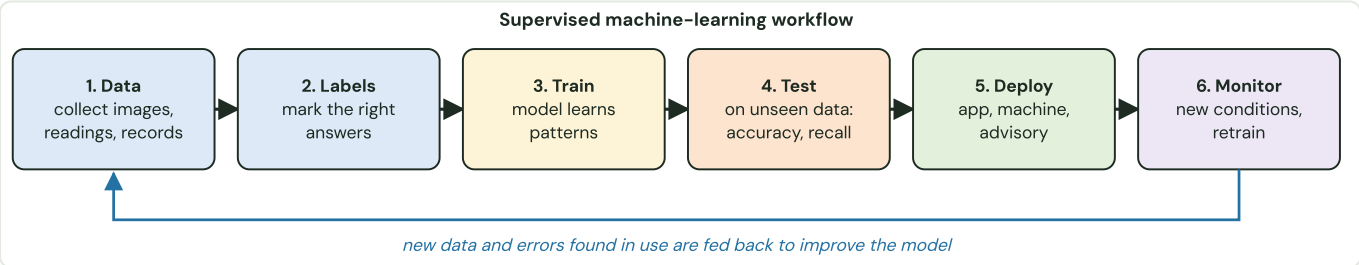


Figure 4. Supervised machine-learning workflow. Original diagram.

A model must be tested on data it has **not** seen in training. A model that memorises its training data but fails on new data is said to **overfit**; one that performs well on new data **generalises**.

A first algorithm: k-nearest neighbours

The **k-nearest-neighbour (k-NN)** classifier labels a new case by majority vote among the k most similar training cases. Features are first scaled so that one with large numbers does not dominate the distance.

Worked example 1 — classifying crop stress with k-NN in Python (Lecture 8)

```
import math

# Training data: (NDVI, canopy minus air temperature in deg C, label) - illustrative field observations
train = [(0.78, -2.0, "healthy"), (0.74, -1.5, "healthy"), (0.81, -2.5, "healthy"), (0.70, -1.0,
"healthy"),
         (0.52,  1.5, "stressed"), (0.48,  2.0, "stressed"), (0.58,  0.5, "stressed"), (0.45,  2.5,
"stressed")]
new_plot = (0.62, 0.0)

# Scale both features to 0-1 so that neither dominates the distance (min-max scaling)
lo = [min(r[i] for r in train) for i in (0, 1)]
hi = [max(r[i] for r in train) for i in (0, 1)]
scale = lambda x, i: (x - lo[i]) / (hi[i] - lo[i])

def distance(a, b):
    return math.sqrt(sum((scale(a[i], i) - scale(b[i], i)) ** 2 for i in (0, 1)))

ranked = sorted(train, key=lambda r: distance(r, new_plot))
k = 3
for r in ranked[:k]:
    print(f"neighbour NDVI={r[0]:.2f}, dT={r[1]:+.1f} -> {r[2]:8s} distance {distance(r,
new_plot):.3f}")
votes = [r[2] for r in ranked[:k]]
print("prediction:", max(set(votes), key=votes.count), votes)

Output:
neighbour NDVI=0.58, dT=+0.5 -> stressed distance 0.149
neighbour NDVI=0.70, dT=-1.0 -> healthy  distance 0.299
neighbour NDVI=0.52, dT=+1.5 -> stressed distance 0.409
prediction: stressed ['stressed', 'healthy', 'stressed']
```

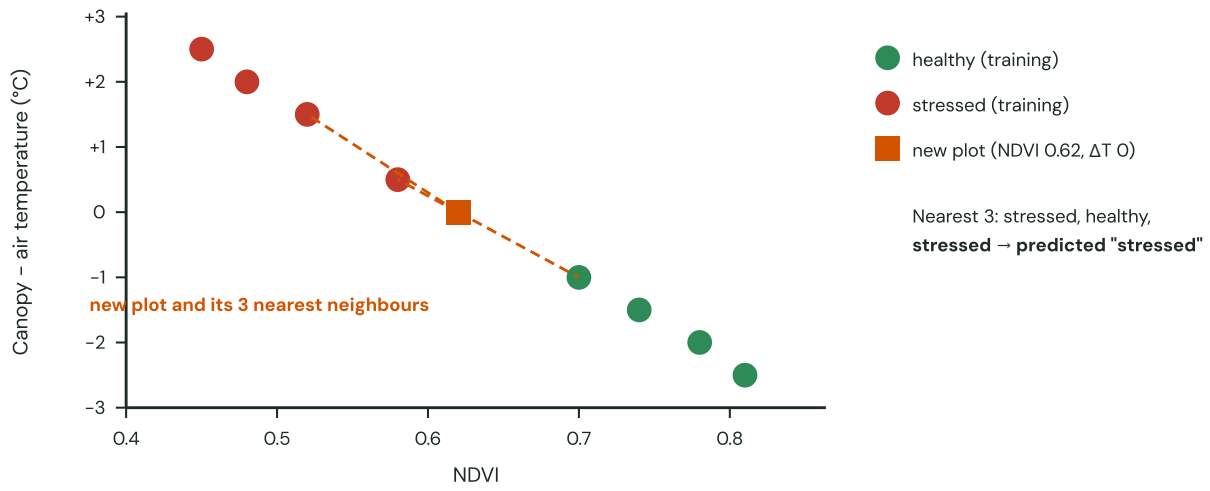


Figure 5. Training plots and the new plot in feature space; two of its three nearest neighbours are stressed. Original chart from illustrative data.

The program was run and the output shown is its actual output. The data are illustrative: a stressed crop tends to have lower NDVI (Lecture 15) and a warmer canopy because it transpires less. Check by hand: scaled NDVI of the new plot = (0.62 − 0.45) ÷ 0.36 = 0.472 and scaled ΔT = (0 + 2.5) ÷ 5 = 0.5; for the nearest neighbour (0.361, 0.6) the distance is √(0.111² + 0.100²) = 0.149.

Judging a classifier

$$\text{Accuracy} = (\text{TP} + \text{TN}) \div \text{total} \qquad \text{Precision} = \text{TP} \div (\text{TP} + \text{FP}) \qquad \text{Recall} = \text{TP} \div (\text{TP} + \text{FN}) \qquad \text{F1} = 2\text{PR} \div (\text{P} + \text{R})$$

Worked example 2 — evaluating an AI pest detector (illustrative test of 200 images: 80 infested, 120 healthy)
True positives (infested, flagged) TP = 68; false negatives (infested, missed) FN = 12; false positives (healthy, flagged) FP = 18; true negatives TN = 102.
Accuracy = (68 + 102) ÷ 200 = 85.0%. Precision = 68 ÷ (68 + 18) = 79.1% — of the images flagged, how many were really infested.
Recall = 68 ÷ (68 + 12) = 85.0% — of the infested images, how many were caught. F1 = 2 × 0.791 × 0.850 ÷ (0.791 + 0.850) = 0.82.
For pest warnings, high recall matters (missing an outbreak is costly); for deciding to spray, precision matters (false alarms waste chemicals).

6. AI in agriculture

A review of machine learning in agricultural production grouped its applications into **crop management** (yield prediction, disease detection, weed detection, crop quality and species recognition), **livestock management** (animal welfare and production), **water management** and **soil management** (Liakos et al., 2018).

Application area	Example	Lecture
Big data and IoT	Sensor streams and farm data for prediction and alerts	23
Autonomous crop and livestock management	Self-driving machines, animal monitoring	24
Crop health and intelligent spraying	Disease and pest detection from images; targeted application	25
Precision agriculture	Yield maps and predictive analytics	26
Weeding and harvesting robots	Vision-guided weeding and picking	27
Sorting and food processing	Grading by size, colour and defects	28
Advisory and services	Multilingual chatbots and photo-based pest advice	14

Case study: deep learning for plant-disease diagnosis

Mohanty, Hughes and Salathé (2016) trained a deep convolutional neural network on 54,306 public images of healthy and diseased leaves of 14 crops with 26 diseases, photographed under controlled conditions. It identified the correct crop–disease class in 99.35% of held-out test images. But on small sets of images from other sources, taken in different conditions, its accuracy fell to about 31% (31.40% and 31.69%) — still far better than the 2.63% of random guessing, yet a clear warning: **a model is only as good as the match between its training data and the real field.**

AI initiatives in India

- **IndiaAI Mission** — approved by the Union Cabinet on 7 March 2024 with an outlay of ₹10,371.92 crore; it includes public AI compute infrastructure of 10,000 or more GPUs built through public–

private partnership, support for indigenous foundation models and AI start-ups (Press Information Bureau [PIB], 2024).

- **Kisan e-Mitra** — an AI chatbot for the PM-KISAN scheme that answers farmers' queries in their own languages, initially in English, Hindi, Odia, Tamil and Bangla (PIB, 2023).
- **National Pest Surveillance System** — AI-assisted pest identification from farmers' photos (Lecture 14).
- **YES-TECH** — technology-based yield estimation for crop insurance using satellite data and models (Lecture 15).

7. Limitations and responsible AI

Issue	Why it matters in agriculture	Good practice
Data mismatch and bias	Models trained on one region, crop variety or camera may fail elsewhere (as the case study shows)	Train and test on local, representative data; report where a model was validated
Explainability	Farmers and officers need reasons before acting	Show why a recommendation was made; keep simple models where they suffice
Privacy and ownership of data	Farm, location and personal data are sensitive (DPDP Act, 2023, Lecture 6)	Consent, minimal collection, secure storage
Access and digital divide	Smartphones, data and skills are unevenly spread (Lecture 10)	Voice, local languages, offline modes, intermediaries
Over-reliance and errors	A wrong pest or dose recommendation can harm crops and the environment	Human oversight; easy reporting of errors; regular updating

8. Key takeaways

- AI builds machines that perceive, learn, reason and act; the OECD defines an AI system as a machine-based system that infers from input how to generate predictions, content, recommendations or decisions.
- AI can be defined by thinking or acting, humanly or rationally; modern AI mostly builds rational agents, and all current AI is narrow.
- The field was named in the 1955 Dartmouth proposal; key steps include Turing (1950), Samuel's machine learning (1959), expert systems, and deep learning since 2012.
- Machine learning is part of AI and deep learning part of machine learning; other branches include vision, language, knowledge-based systems, search, robotics and fuzzy logic.
- An agent perceives through sensors and acts through actuators; PEAS specifies its task.
- Supervised, unsupervised and reinforcement learning differ in what they learn from; models must be tested on unseen data.
- Accuracy, precision, recall and F1 judge classifiers; lab accuracy can fall sharply in the field.
- India is investing through the IndiaAI Mission and deploying AI in services such as Kisan e-Mitra; responsible use needs local data, explanations, privacy and human oversight.

9. Examination questions

Short-answer questions (2 marks)

Q1. Define artificial intelligence.

The field that builds machines able to perform tasks normally requiring human intelligence, such as perceiving, learning, reasoning and acting.

Q2. Name the four approaches to defining AI.

Thinking humanly, acting humanly, thinking rationally and acting rationally.

Q3. Distinguish narrow AI and general AI.

Narrow AI performs specific tasks; general AI would match human ability across all tasks and does not yet exist.

Q4. What was the Dartmouth proposal?

The 1955 proposal by McCarthy, Minsky, Rochester and Shannon for a 1956 summer study of artificial intelligence, which named the field.

Q5. What is an agent? Expand PEAS.

Something that perceives through sensors and acts through actuators; Performance, Environment, Actuators, Sensors.

Q6. Distinguish supervised and unsupervised learning.

Supervised learning uses labelled examples; unsupervised learning finds patterns or groups in unlabelled data.

Q7. What is overfitting?

When a model fits its training data closely but performs poorly on new data.

Q8. How does the k-NN classifier work?

It labels a new case by majority vote among the k most similar training cases.

Q9. Define precision and recall.

Precision = TP/(TP + FP); recall = TP/(TP + FN).

Q10. State two features of the IndiaAI Mission.

Outlay of ₹10,371.92 crore (2024); public AI compute of 10,000 or more GPUs; support for foundation models and start-ups.

Long-answer questions (5–10 marks) — answer outlines

Q1. Define AI and explain the four approaches to it.

OECD definition; four-approach table; rational agents; narrow vs general AI.

Q2. Trace the history of artificial intelligence.

Figure 1 and milestones — Turing, Dartmouth, Samuel, expert systems, AI winters, statistical ML, deep learning, generative AI, OECD and IndiaAI.

Q3. Describe the branches of AI with agricultural examples.

Figure 2; branches table.

Q4. Explain intelligent agents and write the PEAS description of an agricultural robot.

Figure 3; agent, rational agent; PEAS for a weeding robot.

Q5. Explain the types of machine learning and the supervised-learning workflow. Work a k-NN example.

Types table; Figure 4; overfitting; worked example 1 and Figure 5.

Q6. Discuss applications of AI in agriculture, its evaluation and its limitations.

Liakos categories; applications table; Mohanty case study; worked example 2 metrics; Indian initiatives; responsible-AI table.

10. Glossary

Term	Meaning
Agent	A system that perceives through sensors and acts through actuators
AI system	A machine-based system that infers from input how to generate outputs that influence environments
Classification	Predicting a category, e.g. healthy or diseased
Convolutional neural network	A deep network suited to images
Deep learning	Machine learning with many-layered neural networks
F1 score	Harmonic mean of precision and recall
Generalisation	Good performance on new, unseen data
k-NN	k-nearest-neighbour classifier
Label	The correct answer attached to a training example
Machine learning	Improving at a task by learning from data or experience
Narrow AI	AI designed for a specific task
Overfitting	Memorising training data at the cost of performance on new data
PEAS	Performance, Environment, Actuators, Sensors
Precision	Share of positive predictions that are correct
Recall	Share of actual positives that are detected
Regression	Predicting a number, e.g. yield
Reinforcement learning	Learning actions from rewards and penalties
Training data	Examples used to fit a model

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All diagrams, charts and the program are original. Data in the worked examples are illustrative; figures from the case study are from the cited paper.

Practice MCQs

Q1. The OECD describes an AI system as a machine-based system that

(a) stores files
(b) infers from input how to generate outputs such as predictions or decisions
(c) only follows fixed instructions
(d) connects computers to the Internet

Q2. The approach that builds systems to do the right thing given what they perceive is

(a) thinking humanly
(b) acting humanly
(c) acting rationally
(d) thinking emotionally

Q3. The term "artificial intelligence" comes from the

(a) Turing paper of 1950
(b) Dartmouth proposal of 1955
(c) ImageNet contest
(d) OECD in 2023

Q4. Arthur Samuel's checkers program (1959) is an early example of

(a) expert systems
(b) machine learning
(c) robotics
(d) GIS

Q5. Deep learning is a part of

(a) robotics only
(b) machine learning
(c) database management
(d) fuzzy logic only

Q6. In PEAS, the "A" stands for

(a) Accuracy
(b) Actuators
(c) Algorithms
(d) Agents

Q7. Grouping fields into management zones without labels is

(a) supervised learning
(b) unsupervised learning
(c) reinforcement learning
(d) overfitting

Q8. A model that performs well on training data but poorly on new data is

(a) generalising
(b) overfitting
(c) clustering
(d) labelling

Q9. The k-NN classifier predicts the class of a new case by

(a) the average of all data
(b) majority vote among the k most similar cases
(c) a fixed rule
(d) random choice

Q10. Recall is calculated as

(a) TP ÷ (TP + FP)
(b) TP ÷ (TP + FN)
(c) (TP + TN) ÷ total
(d) FP ÷ (FP + TN)

Q11. In Mohanty et al. (2016), accuracy on images from other conditions was about

(a) 99%
(b) 85%
(c) 31%
(d) 3%

Q12. The IndiaAI Mission approved in 2024 has an outlay of about

(a) ₹1,037 crore
(b) ₹10,372 crore
(c) ₹1,00,000 crore
(d) ₹372 crore

Answers

Q1 — (b) infers from input how to generate outputs such as predictions or decisions
Q2 — (c) acting rationally
Q3 — (b) Dartmouth proposal of 1955
Q4 — (b) machine learning
Q5 — (b) machine learning
Q6 — (b) Actuators
Q7 — (b) unsupervised learning
Q8 — (b) overfitting
Q9 — (b) majority vote among the k most similar cases
Q10 — (b) TP ÷ (TP + FN)
Q11 — (c) 31%
Q12 — (b) ₹10,372 crore